CHAPTER 1

MATHEMATICAL EXPECTATION

CHAPTER ESSENCE AT A GLANCE:
Master mathematical expectation of random variables, expectation of linear combinations, covariance, correlation in expectation terms, and Cauchy-Schwarz inequality.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Definition of Expected Value:** Expectation E(X) is the probability-weighted arithmetic average of all possible values of random variable X; represents the long-run center of gravity.

• **Linearity of Expectation:** Expectation is strictly linear: E(aX + bY + c) = a E(X) + b E(Y) + c holds unconditionally whether X and Y are independent or correlated.

• **Multiplication Theorem and Independence:** If random variables X and Y are statistically independent, E(XY) = E(X) * E(Y); Cov(X, Y) = E(XY) - E(X)*E(Y) = 0.

• **Variance and Covariance Properties:** Var(X) = $E(X^2)$ - $[E(X)]^2$; Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov(X, Y); Correlation r(X,Y) = Cov(X,Y) divided by [SD(X) * SD(Y)].

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$E(aX + bY + c) = a\,E(X) + b\,E(Y) + c$$

Universal Additive Linearity of Mathematical Expectation

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)\,E(Y) \Longleftrightarrow X \perp Y \rightarrow \mathrm{Cov}(X, Y) = 0$$

Bivariate Covariance and Independence Invariant

$$\mathrm{Var}(aX \pm bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) \pm 2ab\,\mathrm{Cov}(X, Y)$$

Variance of Linear Bivariate Combinations

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X\,\sigma_Y} \rightarrow -1 \leq \rho \leq +1$$

Expectation Formulation of Correlation Coefficient

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** If X and Y are independent, Cov(X, Y) = 0; but zero covariance does NOT necessarily imply independence (uncorrelated does not mean independent).

• **Alert:** Var(X - Y) = Var(X) + Var(Y) for independent variables; notice the PLUS sign between variances because $(-1)^2$ = +1.

• **Alert:** Expectation E(X) can be negative, zero, or positive, but Variance Var(X) is strictly non-negative.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Mathematical Expectation for: (a) A discrete random variable, (b) A continuous random variable. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Discrete:** E(X) = sum over all x of [x * p(x)], provided the series converges absolutely. **[1 Mark]**

• **(b) Continuous:** E(X) = integral from -infinity to +infinity of [x * f(x) dx], where f(x) is the probability density function. **[1 Mark]**

2. State and prove the Multiplication Theorem of Expectation for two independent random variables X and Y. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Statement:** If X and Y are two independent random variables, then E(XY) = E(X) * E(Y). **[1 Mark]**

• **Proof (Discrete):** E(XY) = sum_x sum_y [x * y * p(x, y)]. Since X and Y are independent, p(x, y) = p(x) * p(y). **[1 Mark]**

• **Factoring:** E(XY) = [sum_x x * p(x)] * [sum_y y * p(y)] = E(X) * E(Y). Proved. **[1 Mark]**

3. Let X and Y be two random variables such that E(X) = 2, E(Y) = 3, $E(X^2)$ = 8, $E(Y^2)$ = 13, and E(XY) = 7. Calculate: (a) Var(X), (b) Var(Y), (c) Cov(X, Y), (d) Correlation coefficient r(X, Y). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **(a) Var(X):** $E(X^2) - [E(X)]^2 = 8 - 2^2 = 8 - 4 = 4$. Standard Deviation SD(X) = 2. **[1 Mark]**

• **(b) Var(Y):** $E(Y^2) - [E(Y)]^2 = 13 - 3^2 = 13 - 9 = 4$. Standard Deviation SD(Y) = 2. **[1 Mark]**

• **(c) Cov(X, Y):** E(XY) - E(X)*E(Y) = 7 - (2 * 3) = 7 - 6 = 1. **[1 Mark]**

• **(d) Correlation r(X, Y):** Cov(X, Y) divided by [SD(X) * SD(Y)] = 1 divided by (2 * 2) = 1 divided by 4 = 0.25. **[1 Mark]**

4. If X and Y are independent random variables with Var(X) = 5 and Var(Y) = 3, find: (a) Var(2X + 3Y), (b) Var(2X - 3Y). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Rule:** For independent variables, Cov(X, Y) = 0. Formula: $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$. **[1 Mark]**

• **(a) Var(2X + 3Y):** 2^2 * Var(X) + 3^2 * Var(Y) = 4 × 5 + 9 × 3 = 20 + 27 = 47. **[1 Mark]**

• **(b) Var(2X - 3Y):** 2^2 * Var(X) + $(-3)^2$ * Var(Y) = 4 × 5 + 9 × 3 = 20 + 27 = 47. (Note: Variances add up). **[1 Mark]**

5. 'If Cov(X, Y) = 0, X and Y must be independent.' Is this statement always true? Justify your answer. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Statement Evaluation:** False. Zero covariance implies lack of linear relationship (uncorrelated), but does not necessarily imply statistical independence. **[1 Mark]**

• **Counter-example:** If $Y = X^2$ and X takes values -1, 0, +1 with equal probability 1 divided by 3, Cov(X, Y) = 0 even though Y is completely determined by X. **[1 Mark]**

6. Prove that for any constant c: (a) E(c) = c, (b) Var(c) = 0. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) E(c):** E(c) = sum c * p(x) = c * sum p(x) = c * 1 = c. **[1 Mark]**

• **(b) Var(c):** $\text{Var}(c) = E[(c - E(c))^2] = E[(c - c)^2] = E[0] = 0$. A constant has zero dispersion. **[1 Mark]**